

Pravan Omprakash

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PhD candidate in Materials Science with expertise in **Atomistic modelling** and **Data-driven modeling**. Experienced in building **Python** tools for large-scale materials discovery using **Density Functional Theory**. Interested in applying **ML** and **statistical analysis** to solve complex data-driven problems.

EDUCATION

2023 - PhD candidate from **Washington University in St Louis, USA** (GPA: 3.75/4)
2018 - 2022 BTech from **National Institute of Technology, Karnataka, India** (GPA: 3.6/4)

RESEARCH EXPERIENCE

PhD Candidate @ [MCUBE Lab](#), WashU Jan 2023 - present

Develop methods to accelerate alloy discovery of High Entropy Alloys using a mix of Density Functional Theory, Classical Thermodynamics and ML, under the guidance of [Dr. Rohan Mishra](#). Provide theoretical support for experimental observations in polar materials like [hexagonal manganites](#) and [ZrO₂](#).

Undergraduate Research Jan 2019 - July 2022

- Worked with [Dr. Kisor Kumar Sahu](#) and Dr. Swayamjyoti S at IIT-Bhubaneswar to develop ML models for inverse design of metamaterials and [vibration-based energy harvesters](#).
- Contributed to developing a U-NET based segmentation model for detecting lung [X-ray image opacities](#) with Dr Avantika Vardhan at [Feinstein Institutes for Medical Research](#).

SELECTED PROJECTS

Exploring the interplay of disorder and order in materials

- **Visualizing high-dimensional spaces using SymPlex plots** • Publication @ [Scripta Materialia](#)
 - Presentation @ MRS Fall'25 (**Awarded Best Oral Presentation**) • Code @  • Poster @ TMS'26
- **Rapid phase diagram prediction of multinary alloys** • Presentation @ MRS Fall'25 & TMS'26
 - Poster @ [NASSCC'25](#)

Investigating ferroelectricity in Hafnia

- **Hole Doping lowers the coercive field of Hafnia** • Publication @ [Physical Review Materials](#)
 - Poster @ Ferro'24
- **Antiferroelectric Phase Stabilization at 2D limit in Hafnia** • Preprint @ [Arxiv](#) (Revision in Science)

Developing ML models for various applications

- **Graph Neural Networks for material property prediction** • Publication @ [Computational Materials Science](#) • Code @ 
- **CNN for facial movement recognition** • Presentation @ [AAAI'21](#) (**Top 20 student abstracts**)
 - Code @ 

TECHNICAL SKILLS

Programming Python, NumPy, Pandas, SciPy, Matplotlib, Plotly
Machine Learning PyTorch, TensorFlow, Scikit-learn

Computational Materials Science Density Functional Theory (VASP), Molecular Dynamics (LAMMPS), Pymatgen